



# Tecnológico de Monterrey

## 4.2.3 Actividad

### Propuesta Completa de Juego

Software Construction and Decision Making

Grupo 501

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# 1. Index

|                        |           |
|------------------------|-----------|
| <b>1. Index</b>        | <b>2</b>  |
| <b>2. Game Design</b>  | <b>2</b>  |
| 2.1 Summary            | 3         |
| 2.2 Gameplay           | 3         |
| 2.3 Mindset            | 4         |
| <b>3. Technical</b>    | <b>5</b>  |
| 3.1 Screens            | 5         |
| 3.2 Controls           | 6         |
| 3.3 Mechanics          | 6         |
| <b>4. Level Design</b> | <b>7</b>  |
| 4.1 Themes             | 7         |
| 4.2 Game Flow          | 11        |
| <b>5. Development</b>  | <b>12</b> |
| 5.1 Abstract Classes   | 12        |
| 5.2 Derived Classes    | 12        |
| <b>6. Graphics</b>     | <b>12</b> |
| 6.1 Style Attributes   | 12        |
| 6.2 Graphics Needed    | 13        |
| <b>7. Sounds/Music</b> | <b>13</b> |
| 7.1 Style Attributes   | 13        |
| 7.2 Sounds Needed      | 14        |
| 7.3 Music Needed       | 14        |
| <b>8. Schedule</b>     | <b>15</b> |

## 2. Game Design

RPG

Roguelike

The map should change (not just the enemies)

It has to include cards

Addition and subtraction (for children)

Inspiration from:

- Plague Inc.
- Card Wars: Adventure Time
- Super Smash Bros.
- Super Mario Bros.

### 2.1 Summary

*Math Smash: Card Adventure* is a strategic roguelike deck-building game designed for children, combining mathematics learning with engaging gameplay mechanics.

Players explore a randomly generated map that changes every run, ensuring that not only enemies but also combats and rewards vary each time. Before each battle, players build and customize their deck, choosing between attack, defense, and skill cards.

The core gameplay revolves around solving math problems under time pressure to activate cards. Different maps introduce increasing levels of mathematical complexity, ranging from basic addition and subtraction, division and multiplication to mixed operations.

The game combines entertainment with learning, helping children improve math skills while playing.

Rather than relying on reflexes, the game emphasizes strategy, planning, and decision-making, making each run a unique challenge that combines learning with entertainment.

### 2.2 Gameplay

The gameplay focuses on strategic planning, adaptation, and progression across a dynamic roguelike structure.

Players navigate through a branching map system that includes battles, chests, and special combats. Each map introduces different enemies, mechanics, and levels of difficulty, requiring players to continuously adapt their strategy. The goal is to defeat all the enemies

and final bosses in order to obtain new cards and progress to other increasingly difficult maps, where the player will need to build better strategies and decks.

Before combat, players select their deck, which consists of:

- Attack cards, activated by solving math problems to deal damage
- Defense cards, used to reduce or avoid incoming damage
- Skill cards, special abilities obtained through winning a combat with the final boss of each map, or by surprise rewards.

During combat, damage and defense cards are activated by solving math problems within a limited time. Our game is designed to be challenging, but not stressful for the children, so in order to keep a balance we made a ‘timer system’. It consists of a time bar which will define the percentage of points provided by the cards depending on the time it takes the player to answer.

Each run presents unique challenges such as the different math problems the player will need to solve, and the limitation of 1 ‘Skills’ type of cards per deck. Along the game there will be random events, which will depend on the type of chest the player finds, there are two types of chests, ‘Reward chest’ and ‘Trap chest’

In order to encourage experimentation and replayability, the team decided that even after losing a combat, players can retain certain progress. For each combat, the Win / Lose system is established as it follows:

- a. If the player wins: Gains 1 new card, the player levels up. If they win against a final boss, they unlock 1 Skill card.
- b. If the player loses: The player loses all normal deck cards, but they will keep every Skill card.

## 2.3 Mindset

*Math Smash: Card Adventure* is designed to foster a strategic, analytical, and confident mindset while maintaining a fun and accessible experience for children.

Target Audience

- Age: 8–14
- Kids learning or looking to improve their math abilities.

Educational Value

The game improves:

- Arithmetic skills
- Quick thinking

- Problem-solving

The game encourages players to:

- Strategic thinking
- Fast decision-making under pressure
- Confidence in math skills

By integrating learning directly into gameplay, the experience feels natural rather than forced. Players are rewarded not only for success, but also for improving their thinking process and adapting their strategies.

The game is designed to make players feel:

- Rewarded for quick thinking
- Motivated to improve
- Curious about each new run

Through randomness, progression, and satisfying feedback, the game creates a balance between challenge, learning, and enjoyment.

## 3. Technical

### 3.1 Screens

For the user to navigate intuitively, the game will include several screens, ranging from the start screen to the character selection screen. These screens are described below:

1. Home Screen
  - Main menu, with the buttons: Play / Credits / Options menu
  - Background music
2. Options Menu
  - Background music settings
  - Effects sounds settings
  - Difficulty preferences (if the player wants to choose a specific type of math operations)

Once they press 'Play' button, the screens will show:

3. Character Selection
  - The player can choose between 3 different skins
4. Instructions Screen. It will explain:
  - Control keys
  - Math system
  - Timer bar
  - Deck and card usage

5. Deck Building Screen
  - Player selects between:
    - Card types
    - Credit levels
  - Preview Deck strategy
6. Main Game Screen
  - Map navigation
  - Combat UI:
    - Cards available in the chosen deck
    - Math problem, with the input to answer
    - Timer bar (green / yellow / red)
    - Player and enemy status or life bar
7. Reward Screen
  - It shows the new card won in the combat.
8. Credits
  - Team members
  - References to graphic elements
  - Background music reference
  - Back to menu button

## 3.2 Controls

Mouse:

- Select cards
- Navigate UI

Keyboard:

- Input math answers
- Confirm actions

## 3.3 Mechanics

The mechanics of the game define how the player interacts with combat, cards, and progression systems. The gameplay combines mathematical problem-solving with a turn-based card system, where performance depends on both accuracy and response time.

### 1. Combat Flow

All actions the main characters can make, depend on solving math problems, therefore the system focuses on rewarding accuracy and speed, so there can be a balance between challenge and accessibility.

Each combat follows a turn-based loop with clearly defined steps:

1. The player selects a card
2. The system generates a math problem
3. The player inputs an answer
4. The system evaluates correctness and response time
5. The card effect is applied (damage or defense)
6. The enemy performs its action, dealing damage to the player

This cycle repeats until either the player defeats all enemies in the world or the player's HP reaches zero.

## 2. Card System

As the player progresses across combats and worlds, for each deck they have the restriction of using only 1 Skill card per deck. They can obtain 3 different card types to play:

- Attack: deals damage to the enemy
- Defense: blocks damage received from the enemy
- Skill: special ability (will be explained below)

## 3. Timer System

The effectiveness of each action depends on response speed:

- Green: 100% effect
- Yellow: 75% effect
- Red: 50% effect
- Wrong answer or timeout: 0% effect

## 4. Chests System

Chests appear throughout the map and provide either rewards or challenges, but in order to make each map unique, there are two types of chests:

- Normal: gives rewards (random cards) or HP (health points)
- Trap: triggers a challenge (math problem) or a combat with a surprise enemy.

If the Trap chest triggers a Surprise Enemy, there are two possibilities:

- a. Card Thief: locks a card until solved
- b. Swapper: randomizes cards

## 5. Game Attributes and Values

### 5.1 Core Attributes

- HP (Health Points): determines the survival of player and enemies
- Attack Power: base damage dealt by enemies
- Defense Effect: percentage of damage reduction when defense is activated

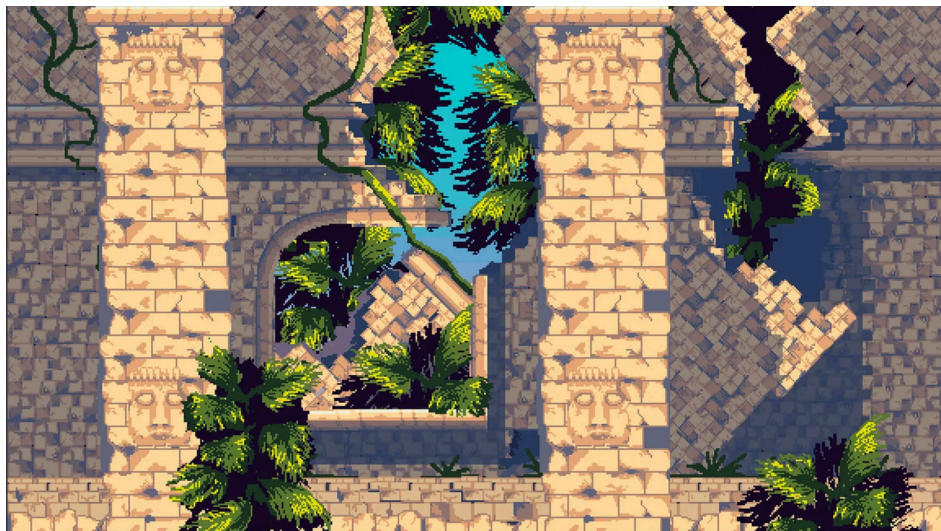
### 5.2 Damage Model

- Damage output depends on:
  - Activation success
  - Timer multiplier
- Damage taken depends on:
  - Enemy attack power
  - Defense effect
  - Timer multiplier

## 4. Level Design

### 4.1 Themes

1. Ancient Temple
  - i. Mood: mysterious, adventurous
  - ii. Objects:
    - a. Ambient: ruins, statues, plants
    - b. Interactive: enemies, chests.



2. Castle
  - i. Mood: tense, dangerous



ii. Objects:

- Ambient: trees, walls
- Interactive: enemies, chests



3. Wasteland

i. Mood: hostile, chaotic

ii. Objects:

- Ambient: broken terrain, rocks.
- Interactive: strong enemies, chests.



## Interactive

The interactive elements for our game are going to be spawned randomly, but there will be a certain amount of interactive objects depending on the category. A world or theme will have two or three normal enemies, 1 final boss, and 1 chest which will have a random interactive event or object, as mentioned before.

Enemies:

- Slimes, golems, spiders, predator plant, skeleton, evil bat
- Each enemy has different behaviors

Bosses:

- Vampire king, Bone Mage, Titan
- Each boss introduces unique mechanics and difficulty

Purpose:

These elements create interaction and decision-making opportunities, encouraging strategic thinking and meaningful risk-reward decisions.

## **Enemy Skills**

Each enemy type is designed to create different challenges and force the player to adapt their strategy.

Basic Enemies:

Slime:

- Simple attack with low damage
- Purpose: Introduce core mechanics to the player
- Skill: “Sticky Hit” – reduces player card effectiveness by 10% for one turn
- 

Golem:

- High defense enemy
- Skill: “Stone Shield” – reduces incoming damage by 50% for one turn
- Purpose: Encourages strategic card usage

Spider:

- Unpredictable enemy
- Skill: “Web Trap” – disables one random card for the next turn
- Purpose: Forces adaptability and planning

Predator plant:

- Fast attack enemy
- Skill: “Quick Strike” – attacks before the player completes their action if time is low
- Purpose: Adds pressure to the timer system

Special Enemies (from Trap Chests):

Card Thief:

- Skill: “Steal Card” – locks one card until the player answers correctly

Swapper:

- Skill: “Chaos Swap” – replaces one card with a random one temporarily

Bosses:

Vampire king:

- Fast and aggressive
- Skill: “Royal Command” – performs two actions in one turn

Bone Mage:

- Balanced boss
- Skill: “Double Action” – increases enemy damage for one turn

Titan:

- High HP boss
- Skill: “Earth Smash” – deals heavy damage but has a delay
- Purpose: Tests endurance and consistency

Enemy skills are designed to interact with:

- Timer system
- Card system
- Player decision-making

## Card types and functions:

Skill Cards:

- “Second chance”: gives the player the option to retry a failed operation.
- “Freeze time”: pauses time for 4 seconds.
- “Clear mind”: the next card (attack or defense) does not require activation (no answer needed).
- “Double power”: doubles the points of any cards.
- “Vitality boost”: Gives the player 10% more life.

Attack and Defense cards:

| Type/Map | Ancient temple                                                                                                      | Castle                                                                                                                       | Wasteland                                                                                             |
|----------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Defense  | <ol style="list-style-type: none"> <li>Dodge: Reduces damage based on the result</li> <li>Shield. Blocks</li> </ol> | <ol style="list-style-type: none"> <li>Barrier defense. Blocks damage equal to the result <math>\times 1.5</math></li> </ol> | <ol style="list-style-type: none"> <li>Counterattack. Returns damage if you answer quickly</li> </ol> |

|        |                            |                                   |                                          |
|--------|----------------------------|-----------------------------------|------------------------------------------|
|        | damage equal to the result |                                   |                                          |
| Attack | a. Quick strike<br>b. Kick | a. Precise lightning<br>b. Poison | a. Lightning explosion<br>b. Fire attack |

## Challenges

Players face different types of challenges:

- Limited time to answer math problems
- Increasing complexity of operations
- Stronger enemies
- Random events (trap chests)
- Limited deck size

These challenges force the player to think strategically and adapt.

## 4.2 Game Flow

1. Player selects character
2. Player enters map
3. Player chooses path
4. Player encounters:
  - Battle
  - Chests
5. Player selects their deck for a combat
6. Combat:
  - They choose a card, solve a math problem, then gain credit according to the Timer bar, and then choose other cards from their deck and so on.
7. Player gains rewards
8. Progress through map
9. Repeats until final boss
10. Fight boss
11. End of run (win / lose → retain some progress → restart)
12. They reach the goal or progression, and the cycle repeats

## 5. Development

### 5.1 Abstract Classes

- BaseEntity
- BaseCard
- BaseEnemy
- BaseMap
- BaseSystem (timer, math)

### 5.2 Derived Classes

The game is built using modular components:

- Player handles deck and stats
- Cards define actions (attack / defense / skill)
- Enemies define behaviors
- Map controls progression
- Timer system controls difficulty

This modular design allows easy expansion of content.

## 6. Graphics

### 6.1 Style Attributes

The game uses a pixel-art style with bright and colorful visuals to appeal to younger audiences. The shapes must be clear and understandable, with high contrast between the characters and the background. Regarding the UI design, the interface of the game must show a clear separation between game elements (cards, math problems, HP bars) so the player doesn't confuse, and the text must be large and readable for children, with a clear interface.

As our audience is children, we need to ensure clarity and engagement, enhancing both gameplay readability and overall player experience, that's why we decided to go for this style of graphics.

### 6.2 Graphics Needed

1. Card animations (attack, defense, skills)
2. Main character options/skins:

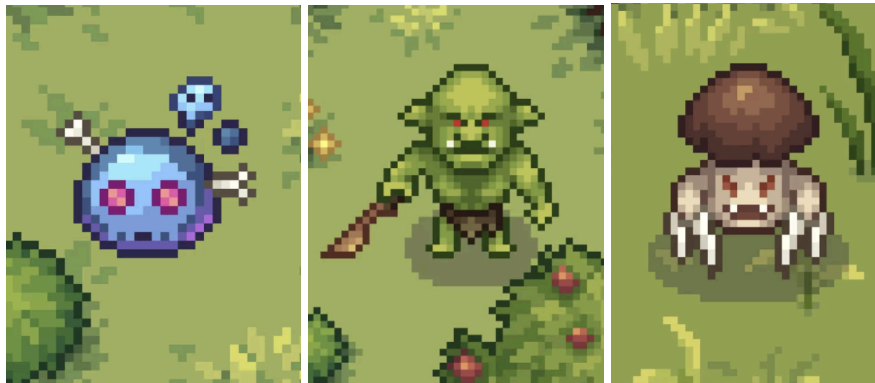


a.

b.

c.

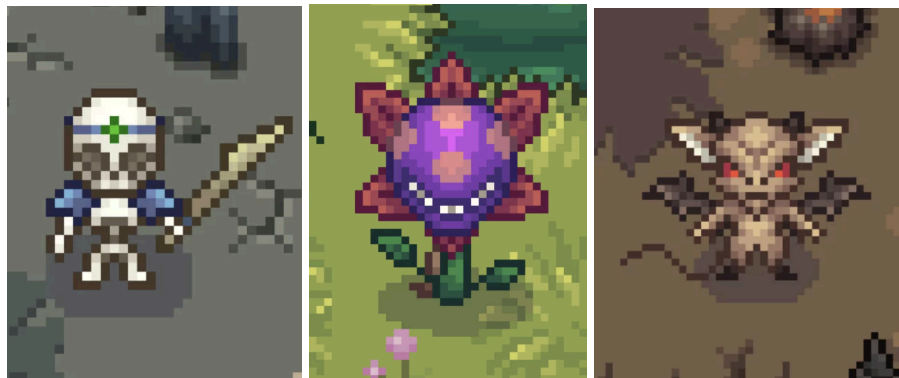
### 3. Basic enemy animations



a. Slime

b. Golem

c. Spider



d. Skeleton

e. Predator plant

f. Evil bat

### 4. Final bosses animation

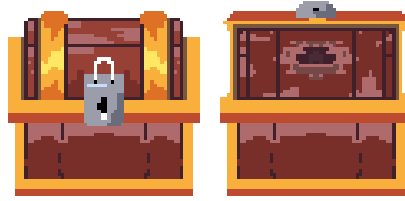


a. Vampire king

b. Titan

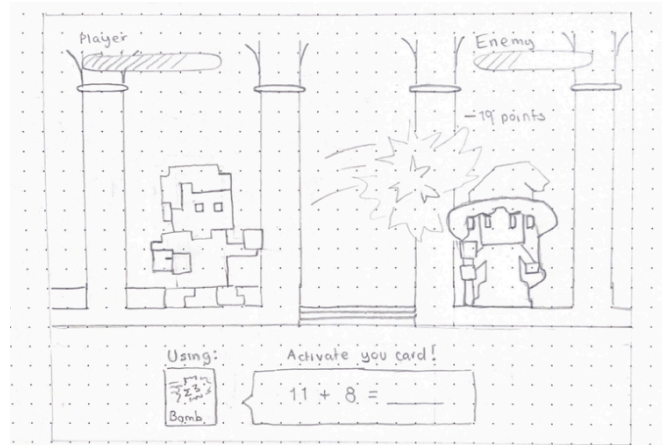
c. Bone Mage

### 5. Chest animation



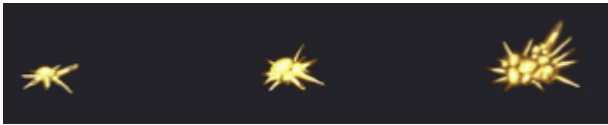
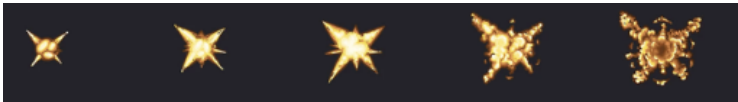
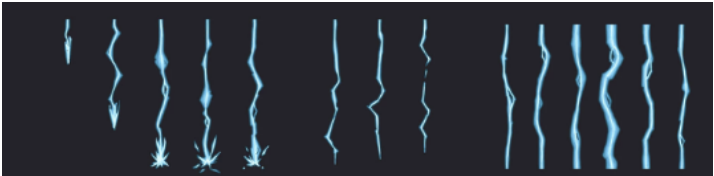
## 6. UI elements (HP bars, timer)

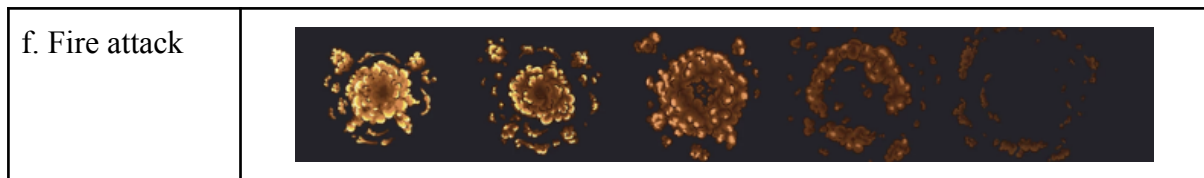
Initial sketch:



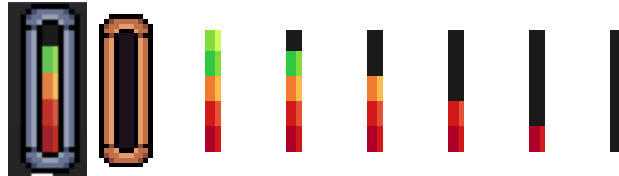
Visual Feedback:

### 1. Animations when attacking or defending

|                        |                                                                                      |
|------------------------|--------------------------------------------------------------------------------------|
| a. Quick strike        |  |
| b. Simple kick         |  |
| c. Precise lightning   |  |
| d. Poison              |  |
| e. Lightning explosion |  |



2. Effects for correct and incorrect answers
3. Timer color changes (green, yellow, red)



4. Life bars:



## 7. Sounds/Music

### 7.1 Style Attributes

The sound design focuses on clarity and feedback while maintaining a fun and light atmosphere.

Sound Effects:

- Short and clear sounds
- Digital and playful tones
- Immediate feedback for player actions

Music Style:

- Light electronic or playful background music
- Calm during exploration
- More intense during combat and boss fights

Purpose:

This audio design enhances immersion while supporting player focus, reinforcing feedback, and improving the overall gameplay experience.



## 7.2 Sounds Needed

- Card activation
- Correct answer
- Wrong answer
- Enemy attacks
- Victory / defeat sounds at the end of the combat

## 7.3 Music Needed

- Menu theme
- Battle theme (with a common enemy)
- Boss theme
- Exploration theme (while the player is not in combat)

# 8. Schedule

The development of the game is divided into phases:

### Week 5 – Core System Development

- Implement base classes (Player, Card, Enemy, Map)
- Develop basic combat system
- Implement math problem system

### Week 6 – Game Mechanics

- Add card system (attack, defense, skills)
- Implement timer system
- Add enemy behaviors

### Week 7 – Map and Progression

- Develop map navigation system
- Implement nodes (battle, chest, event)
- Add progression system

### Week 8 – Visual and Audio Design

- Design UI and visual elements
- Add animations and effects
- Implement sound effects and music

### Week 9 – Testing and Polishing

- Fix bugs
- Balance gameplay

- Improve user experience

This schedule ensures a structured development process from core systems to polishing and testing.